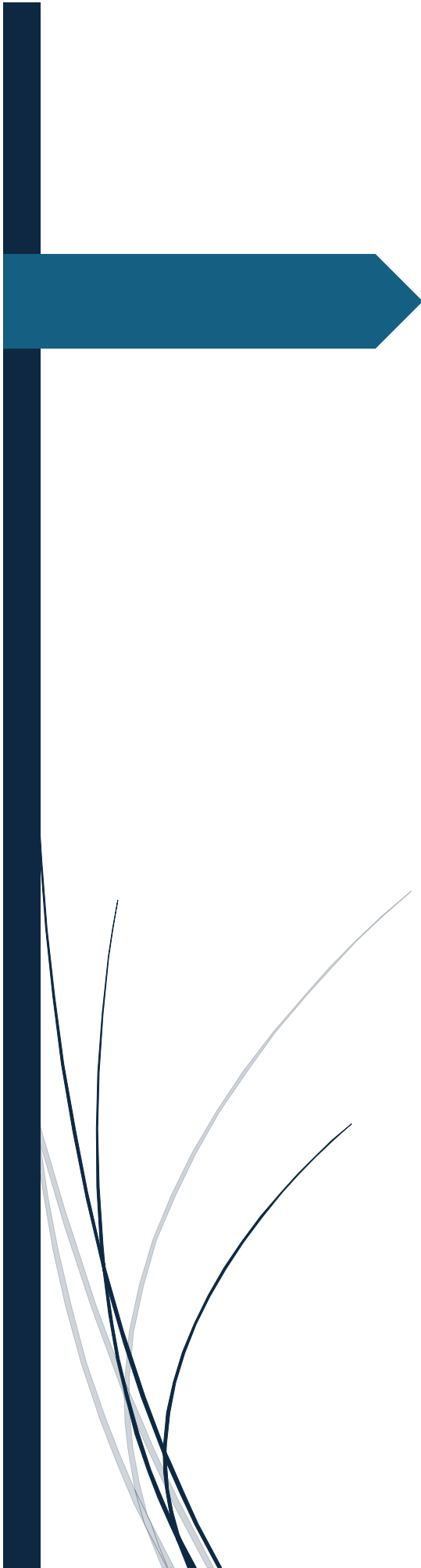




Assignment 4

Network cable Types & Tools
Theory

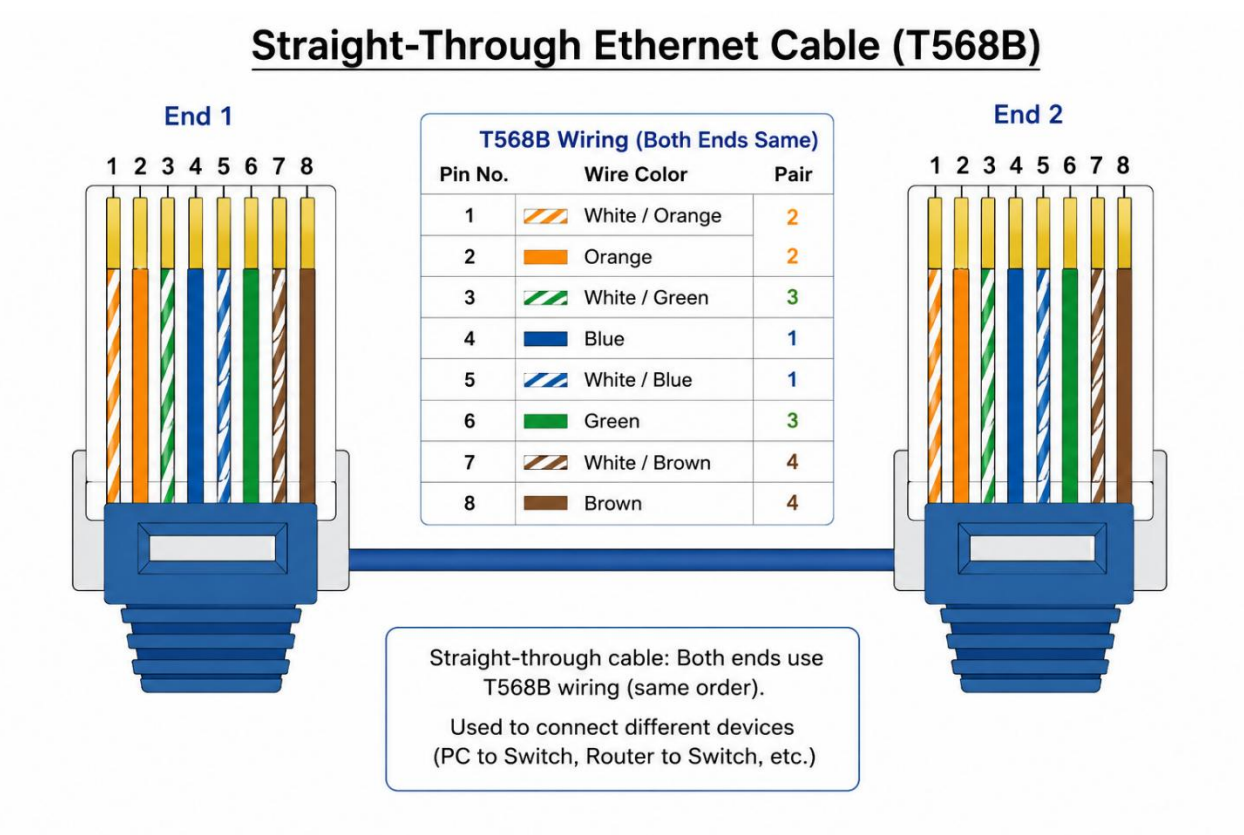
1. List 3 key differences between Ethernet cables and fiber optic cables, and give one real-world example where each type would be used (for example, home WiFi router, data center, college campus backbone).

Difference	Ethernet Cable	Fiber Optic Cable
1. Material	Uses copper wires	Uses glass/plastic fibers
2. Speed & Distance	Good speed, but works for shorter distances	Very high speed and works over long distances
3. Signal	Uses electrical signals	Uses light signals

Real-world examples:

-  **Ethernet:** Connecting a computer to a home Wi-Fi router.
-  **Fiber Optic:** Connecting buildings in a college campus or data center.

2. Draw and label a diagram showing the internal wiring of a straight-through Ethernet cable (use T568B wiring standard), indicating the color of each wire and its pin number.



3. Watch a YouTube video demonstrating how to use an RJ45 crimper, then write a step-by-step guide (in your own words) on how to attach an RJ45 connector to a CAT6 cable.

1. **Cut the cable** to the required length.
2. **Remove the outer cover** of the CAT6 cable.
3. **Separate and straighten the 8 wires.**
4. **Arrange the wires** in T568B order:
 - White-Orange,
 - Orange,
 - White-Green,
 - Blue,
 - White-Blue,
 - Green,
 - White-Brown,
 - Brown.
5. **Cut the wire ends evenly.**
6. **Put the wires into the RJ45 connector** in the same order.
7. **Put the connector into the crimping tool** and press it firmly.
8. **Check the cable** using a cable tester if available

4. Imagine you are setting up a gaming café with 10 PCs for LAN gaming. Decide which cable type (Ethernet or fiber optics) and which tool (RJ45 crimper or fiber cleaver) you would use, and explain your choices in 3-4 sentences. Hint: Consider speed, distance, and budget for your answer.

➤ For a gaming café with **10 PCs**, I would use **Ethernet cables** because they are fast enough for LAN gaming and work well over normal room distances. I would use an **RJ45 crimper** to attach the connectors to the Ethernet cables. Ethernet is also **cheaper and easier to install** than fiber optics. Fiber would be better for very long distances, but it is not necessary for a small gaming café.